No.	Status	Date
HIS-2006-0412-DS-4-LA-102	Issued	2006-02-24

UT Thickness Data Sheet - 4-LA-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-LA-102, a lean amine supply line located within the ATU-100. The inspection was conducted as part of the 10-year full-unit survey detailed in HIS-2006-0412.

Kevin Doucette, acting as the lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Specifications

Line 4-LA-102 is constructed from A106 Gr.B with an outer diameter of 114.3 mm and a nominal thickness of 6.02 mm. The design limits for this line are 2413 kPa(g) and 93 °C.

The minimum allowable thickness (t-min) for this asset is 3.18 mm. Any reading recorded below this threshold is subject to an NCR.

Procedural Compliance

Thickness measurements were performed in accordance with QSCP-NDT-001 Rev 0, the site NDE procedure issued for ATU-100 pre-startup and in-service NDE.

The inspection also adhered to HIS-UT-002 Rev 1, the ultrasonic thickness measurement procedure submitted by Halward Inspection Services Ltd. for QSCP acceptance, and the requirements of ASTM E797.

Asset Integrity Analysis

The remaining life of 4-LA-102 is calculated by dividing the difference between the current thickness and the t-min of 3.18 mm by the determined corrosion rate. The corrosion rate is the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart.

Based on the Class 2 designation for amine service, the maximum interval for thickness measurements is 10 years. The next reading is due at no more than half the remaining life or within the class maximum.

Thickness readings, measured 2006-02-24

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2001-03-02	6.82	6.59	0.053	0.046	>50	2011-02-24	
6-SG-101-02	7.11	3.56	2001-03-02	6.71	6.51	0.054	0.040	>50	2011-02-24	
6-SG-101-03	7.11	3.56	2001-03-02	6.74	6.42	0.066	0.064	43.2	2011-02-24	
6-SG-101-04	7.11	3.56	2001-03-02	6.73	6.28	0.081	0.090	30.1	2011-02-24	
6-SG-101-05	7.11	3.56	2001-03-02	6.68	6.22	0.092	0.092	28.8	2011-02-24	
6-SG-101-06	7.11	3.56	2001-03-02	6.87	6.49	0.058	0.076	38.4	2011-02-24	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2001-03-02	6.72	6.26	0.088	0.092	29.2	2011-02-24	
6-SG-102-02	7.11	3.56	2001-03-02	6.72	6.25	0.086	0.094	28.5	2011-02-24	
6-SG-102-03	7.11	3.56	2001-03-02	6.69	6.26	0.078	0.086	31.3	2011-02-24	
6-SG-102-04	7.11	3.56	2001-03-02	6.64	6.22	0.077	0.084	31.6	2011-02-24	
6-SG-102-05	7.11	3.56	2001-03-02	6.93	6.72	0.041	0.042	>50	2011-02-24	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	1996-03-08	8.20	7.89	0.031	0.031	>50	2016-02-24	
8-TG-101-02	8.18	4.06	1996-03-08	8.20	7.75	0.045	0.045	>50	2016-02-24	
8-TG-101-03	8.18	4.06	1996-03-08	8.23	7.79	0.044	0.044	>50	2016-02-24	
8-TG-101-04	8.18	4.06	1996-03-08	8.10	7.76	0.034	0.034	>50	2016-02-24	
8-TG-101-05	8.18	4.06	1996-03-08	8.22	7.95	0.027	0.027	>50	2016-02-24	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	1996-03-08	5.99	5.51	0.048	0.048	48.4	2016-02-24	
4-LA-101-02	6.02	3.18	1996-03-08	6.01	5.76	0.025	0.025	>50	2016-02-24	
4-LA-101-03	6.02	3.18	1996-03-08	5.96	5.76	0.020	0.020	>50	2016-02-24	
4-LA-101-04	6.02	3.18	1996-03-08	6.07	5.78	0.029	0.029	>50	2016-02-24	
4-LA-101-05	6.02	3.18	1996-03-08	6.01	5.81	0.020	0.020	>50	2016-02-24	
4-LA-101-06	6.02	3.18	1996-03-08	6.06	5.64	0.042	0.042	>50	2016-02-24	
4-LA-101-07	6.02	3.18	1996-03-08	5.99	5.57	0.042	0.042	>50	2016-02-24	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	1996-03-08	6.02	5.63	0.039	0.039	>50	2016-02-24	
4-LA-102-02	6.02	3.18	1996-03-08	6.07	5.67	0.040	0.040	>50	2016-02-24	
4-LA-102-03	6.02	3.18	1996-03-08	6.07	5.55	0.052	0.052	45.4	2016-02-24	
4-LA-102-04	6.02	3.18	1996-03-08	5.99	5.34	0.065	0.065	33.1	2016-02-24	
4-LA-102-05	6.02	3.18	1996-03-08	6.00	5.62	0.038	0.038	>50	2016-02-24	
4-LA-102-06	6.02	3.18	1996-03-08	6.00	5.83	0.017	0.017	>50	2016-02-24	
4-LA-102-07	6.02	3.18	1996-03-08	6.00	5.63	0.037	0.037	>50	2016-02-24	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	1996-03-08	7.02	6.75	0.027	0.027	>50	2016-02-24	
6-LA-103-02	7.11	3.56	1996-03-08	7.19	6.87	0.032	0.032	>50	2016-02-24	
6-LA-103-03	7.11	3.56	1996-03-08	7.08	6.65	0.043	0.043	>50	2016-02-24	
6-LA-103-04	7.11	3.56	1996-03-08	7.17	6.52	0.065	0.065	45.4	2016-02-24	
6-LA-103-05	7.11	3.56	1996-03-08	7.19	6.49	0.070	0.070	41.7	2016-02-24	
6-LA-103-06	7.11	3.56	1996-03-08	7.16	6.42	0.074	0.074	38.5	2016-02-24	
6-LA-103-07	7.11	3.56	1996-03-08	7.18	6.76	0.042	0.042	>50	2016-02-24	
6-LA-103-08	7.11	3.56	1996-03-08	7.04	6.79	0.025	0.025	>50	2016-02-24	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2001-03-02	8.39	8.27	0.025	0.024	>50	2016-02-24	
4-RA-101-02	8.56	3.81	2001-03-02	8.45	8.27	0.027	0.036	>50	2016-02-24	
4-RA-101-03	8.56	3.81	2001-03-02	8.42	8.10	0.051	0.064	>50	2016-02-24	
4-RA-101-04	8.56	3.81	2001-03-02	8.37	8.34	0.029	0.006	>50	2016-02-24	
4-RA-101-05	8.56	3.81	2001-03-02	8.42	8.29	0.031	0.026	>50	2016-02-24	
4-RA-101-07	8.56	3.81	2001-03-02	8.51	8.30	0.030	0.042	>50	2016-02-24	
4-RA-101-08	8.56	3.81	2001-03-02	8.30	8.07	0.061	0.046	>50	2016-02-24	
4-RA-101-09	8.56	3.81	2001-03-02	8.43	8.43	0.014	0.000	>50	2016-02-24	
4-RA-101-10	8.56	3.81	2001-03-02	8.48	8.27	0.040	0.042	>50	2016-02-24	
4-RA-101-11	8.56	3.81	2001-03-02	8.39	8.14	0.054	0.050	>50	2016-02-24	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2001-03-02	8.48	8.41	0.013	0.014	>50	2016-02-24	
4-RA-102-02	8.56	3.81	2001-03-02	8.50	8.15	0.042	0.070	>50	2016-02-24	
4-RA-102-03	8.56	3.81	2001-03-02	8.37	8.27	0.029	0.020	>50	2016-02-24	
4-RA-102-04	8.56	3.81	2001-03-02	8.35	8.24	0.030	0.022	>50	2016-02-24	
4-RA-102-05	8.56	3.81	2001-03-02	8.36	8.19	0.036	0.034	>50	2016-02-24	
4-RA-102-06	8.56	3.81	2001-03-02	8.38	8.15	0.036	0.046	>50	2016-02-24	
4-RA-102-07	8.56	3.81	2001-03-02	8.49	8.29	0.029	0.040	>50	2016-02-24	
4-RA-102-08	8.56	3.81	2001-03-02	8.36	8.05	0.044	0.062	>50	2016-02-24	
4-RA-102-09	8.56	3.81	2001-03-02	8.35	8.18	0.037	0.034	>50	2016-02-24	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2001-03-02	10.82	10.76	0.024	0.012	>50	2016-02-24	
6-RA-103-02	10.97	4.57	2001-03-02	10.69	10.50	0.059	0.038	>50	2016-02-24	
6-RA-103-03	10.97	4.57	2001-03-02	10.84	10.67	0.032	0.034	>50	2016-02-24	
6-RA-103-04	10.97	4.57	2001-03-02	10.59	10.40	0.058	0.038	>50	2016-02-24	
6-RA-103-05	10.97	4.57	2001-03-02	10.61	10.34	0.061	0.054	>50	2016-02-24	
6-RA-103-06	10.97	4.57	2001-03-02	10.86	10.74	0.027	0.024	>50	2016-02-24	
6-RA-103-07	10.97	4.57	2001-03-02	10.79	10.66	0.033	0.026	>50	2016-02-24	
6-RA-103-08	10.97	4.57	2001-03-02	10.83	10.58	0.039	0.050	>50	2016-02-24	
6-RA-103-09	10.97	4.57	2001-03-02	10.85	10.67	0.034	0.036	>50	2016-02-24	
6-RA-103-10	10.97	4.57	2001-03-02	10.78	10.27	0.075	0.102	>50	2016-02-24	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2001-03-02	7.26	6.98	0.061	0.056	>50	2011-02-24	
3-RV-101-02	7.62	3.56	2001-03-02	7.31	6.93	0.064	0.076	44.2	2011-02-24	
3-RV-101-03	7.62	3.56	2001-03-02	7.15	6.57	0.095	0.116	25.9	2011-02-24	
3-RV-101-04	7.62	3.56	2001-03-02	7.18	6.65	0.094	0.106	29.1	2011-02-24	
3-RV-101-05	7.62	3.56	2001-03-02	7.25	7.01	0.066	0.048	>50	2011-02-24	
3-RV-101-06	7.62	3.56	2001-03-02	7.20	6.74	0.085	0.092	34.4	2011-02-24	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2001-03-02	5.29	4.85	0.068	0.088	23.3	2011-02-24	
2-RV-102-02	5.54	2.79	2001-03-02	5.12	4.84	0.075	0.056	27.2	2011-02-24	
2-RV-102-03	5.54	2.79	2001-03-02	5.24	5.14	0.034	0.020	>50	2011-02-24	
2-RV-102-04	5.54	2.79	2001-03-02	5.33	5.07	0.051	0.052	43.7	2011-02-24	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	1996-03-08	5.48	5.48	0.000	0.000	>50	2016-02-24	
3-RF-101-02	5.49	2.41	1996-03-08	5.45	5.37	0.008	0.008	>50	2016-02-24	
3-RF-101-03	5.49	2.41	1996-03-08	5.45	5.45	0.000	0.000	>50	2016-02-24	
3-RF-101-04	5.49	2.41	1996-03-08	5.46	5.33	0.013	0.013	>50	2016-02-24	
3-RF-101-05	5.49	2.41	1996-03-08	5.45	5.40	0.005	0.005	>50	2016-02-24	
3-RF-101-06	5.49	2.41	1996-03-08	5.52	5.40	0.012	0.012	>50	2016-02-24	
3-RF-101-07	5.49	2.41	1996-03-08	5.42	5.35	0.007	0.007	>50	2016-02-24	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	1996-03-08	3.92	3.85	0.007	0.007	>50	2016-02-24	
2-RF-102-02	3.91	2.16	1996-03-08	3.88	3.88	0.000	0.000	>50	2016-02-24	
2-RF-102-03	3.91	2.16	1996-03-08	3.93	3.80	0.013	0.013	>50	2016-02-24	
2-RF-102-04	3.91	2.16	1996-03-08	3.87	3.81	0.006	0.006	>50	2016-02-24	
2-RF-102-07	3.91	2.16	1996-03-08	3.96	3.85	0.011	0.011	>50	2016-02-24	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	1996-03-08	6.07	5.75	0.032	0.032	>50	2016-02-24	
4-LA-104-02	6.02	3.18	1996-03-08	6.03	5.56	0.047	0.047	>50	2016-02-24	
4-LA-104-03	6.02	3.18	1996-03-08	5.96	5.74	0.022	0.022	>50	2016-02-24	
4-LA-104-04	6.02	3.18	1996-03-08	6.06	5.86	0.020	0.020	>50	2016-02-24	
4-LA-104-05	6.02	3.18	1996-03-08	6.04	5.51	0.053	0.053	43.8	2016-02-24	
4-LA-104-06	6.02	3.18	1996-03-08	6.10	5.83	0.027	0.027	>50	2016-02-24	
4-LA-104-07	6.02	3.18	1996-03-08	6.01	5.62	0.039	0.039	>50	2016-02-24	
4-LA-104-08	6.02	3.18	1996-03-08	5.94	5.50	0.044	0.044	>50	2016-02-24	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2001-03-02	8.40	8.16	0.035	0.048	>50	2016-02-24	
4-RA-104-02	8.56	3.81	2001-03-02	8.41	8.23	0.032	0.036	>50	2016-02-24	
4-RA-104-03	8.56	3.81	2001-03-02	8.46	8.30	0.024	0.032	>50	2016-02-24	
4-RA-104-04	8.56	3.81	2001-03-02	8.35	8.21	0.038	0.028	>50	2016-02-24	
4-RA-104-05	8.56	3.81	2001-03-02	8.29	8.14	0.046	0.030	>50	2016-02-24	
4-RA-104-06	8.56	3.81	2001-03-02	8.39	8.09	0.050	0.060	>50	2016-02-24	
4-RA-104-07	8.56	3.81	2001-03-02	8.29	7.97	0.062	0.064	>50	2016-02-24	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	1996-03-08	3.93	3.64	0.029	0.029	46.4	2016-02-24	
2-ST-101-02	3.91	2.29	1996-03-08	3.90	3.67	0.023	0.023	>50	2016-02-24	
2-ST-101-03	3.91	2.29	1996-03-08	3.94	3.82	0.012	0.012	>50	2016-02-24	
2-ST-101-04	3.91	2.29	1996-03-08	3.82	3.77	0.005	0.005	>50	2016-02-24	
2-ST-101-05	3.91	2.29	1996-03-08	3.91	3.72	0.019	0.019	>50	2016-02-24	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	1996-03-08	5.42	5.13	0.029	0.029	>50	2016-02-24	
3-CW-101-02	5.49	2.79	1996-03-08	5.41	5.20	0.021	0.021	>50	2016-02-24	
3-CW-101-03	5.49	2.79	1996-03-08	5.53	5.08	0.045	0.045	>50	2016-02-24	
3-CW-101-04	5.49	2.79	1996-03-08	5.46	5.38	0.008	0.008	>50	2016-02-24	
3-CW-101-05	5.49	2.79	1996-03-08	5.48	5.33	0.015	0.015	>50	2016-02-24	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	1996-03-08	5.50	5.20	0.030	0.030	>50	2016-02-24	
3-CW-102-02	5.49	2.79	1996-03-08	5.51	5.02	0.049	0.049	45.4	2016-02-24	
3-CW-102-03	5.49	2.79	1996-03-08	5.53	5.07	0.046	0.046	49.4	2016-02-24	
3-CW-102-04	5.49	2.79	1996-03-08	5.51	5.03	0.048	0.048	46.5	2016-02-24	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-24	_____